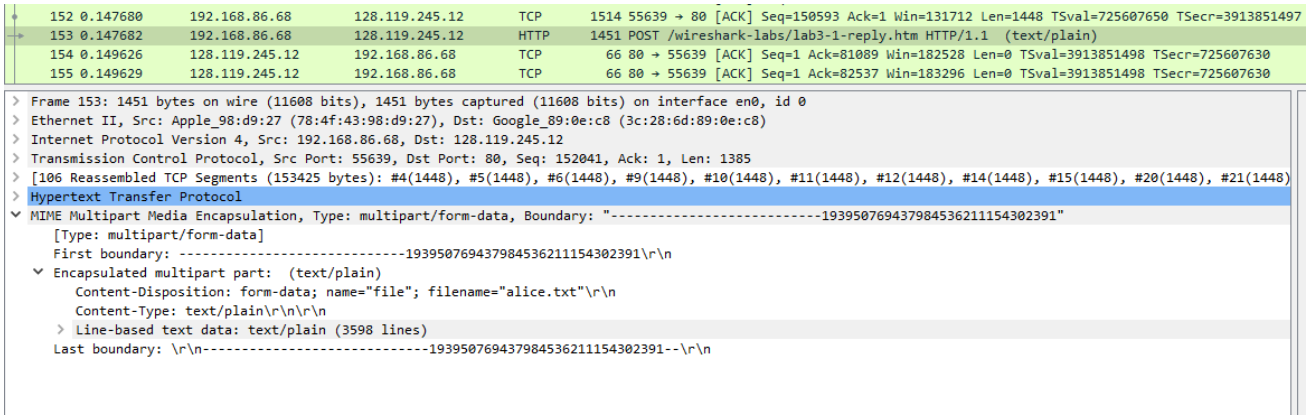
CSE4074 Computer Networks
Homework III

NOTE

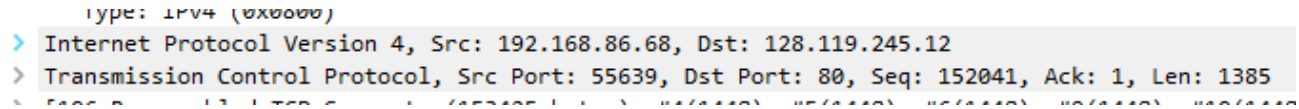
As written in the homework I done it by using Wireshark captured packet file.

> Answer the following questions[3], either from your own live trace, or by opening the Wireshark captured packet file *tcp-wireshark-trace1-1* in http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces-8.1.zip

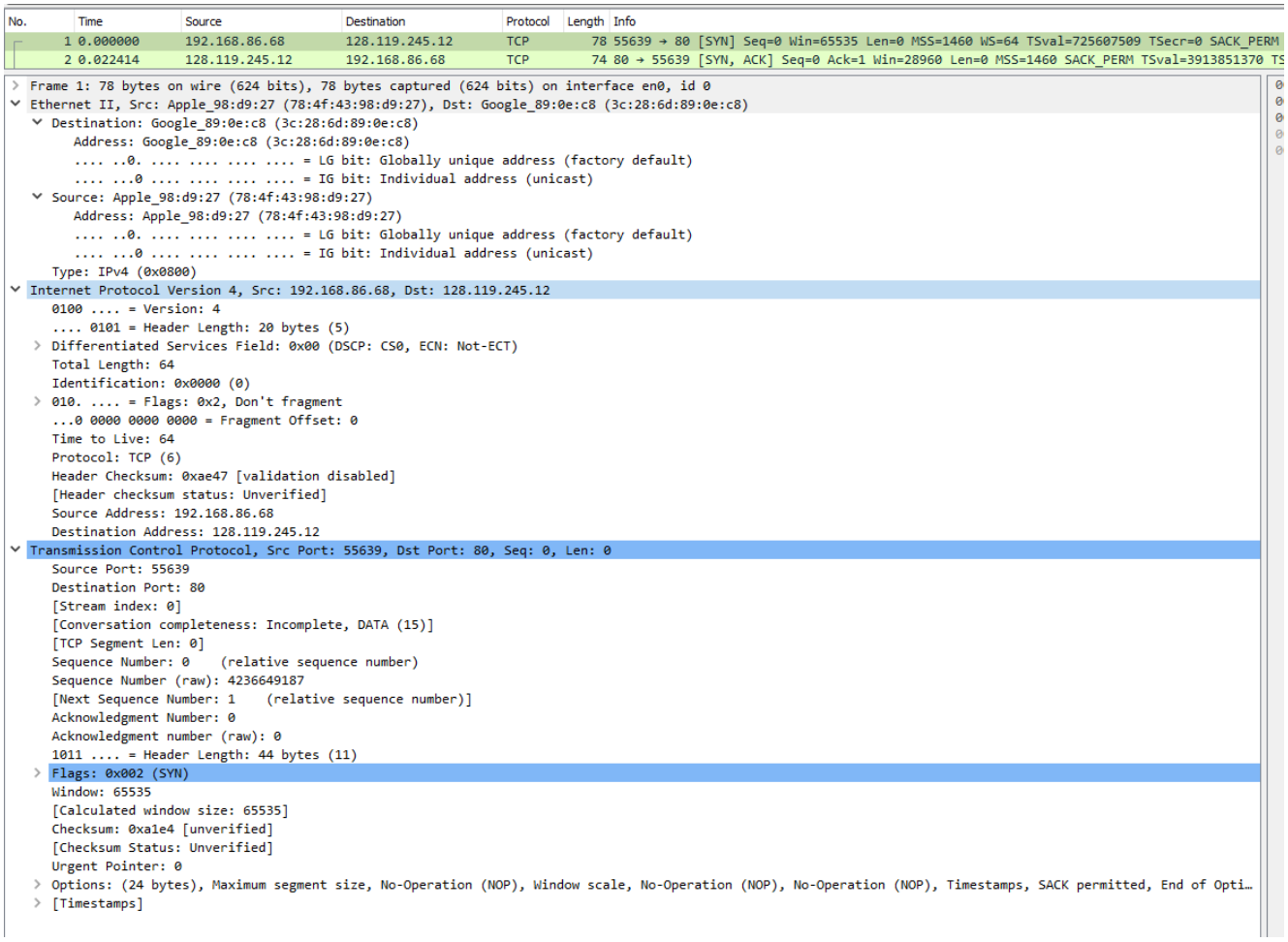
1. 192.168.86.68



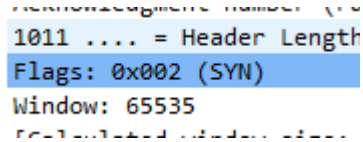
2. 128.119.245.12, port number 80



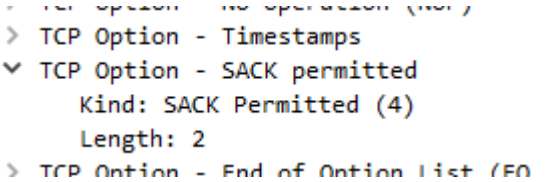
3. The sequence number is 0.



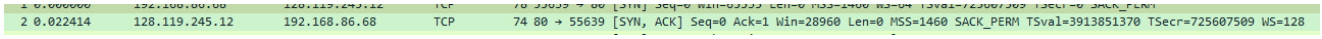
There is a SYN flag to identify this segment as SYN segment.



Yes, selective acknowledgment is enabled.



4. The sequence number is 0.

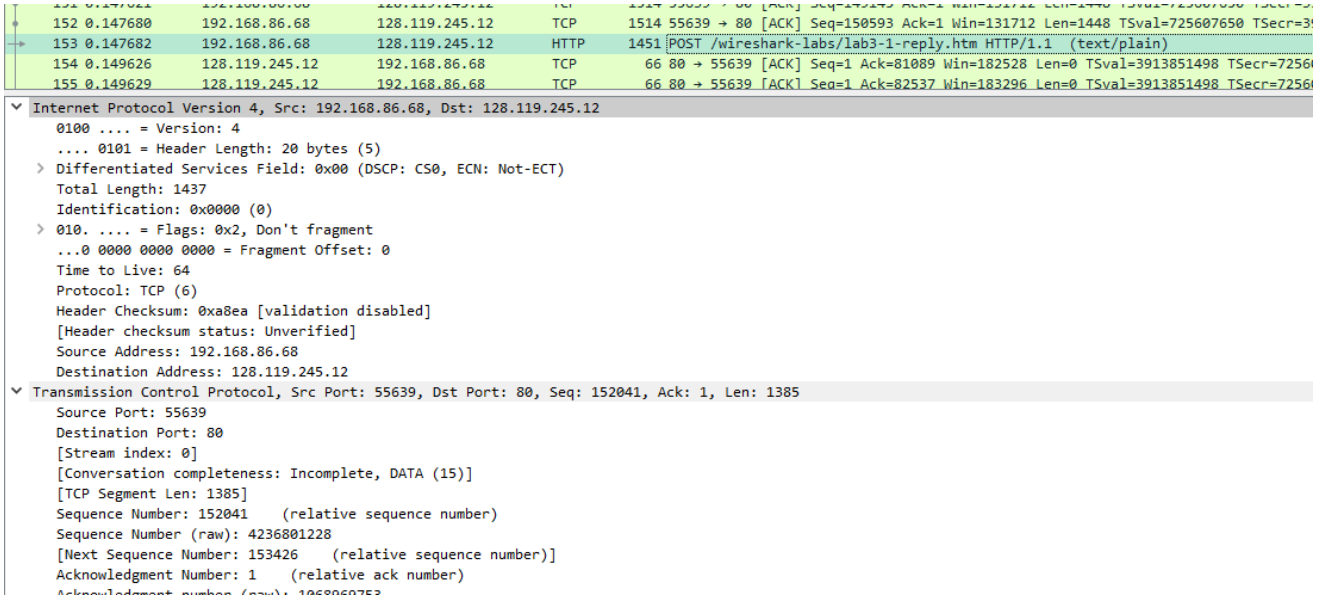


There are SYN, ACK flags

```
1010 .... = Header Length: 40 bytes (10)
Flags: 0x012 (SYN, ACK)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Accurate ECN: Not set
    .... 0... .... = Congestion Window Reduced: Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... ...1 .... = Acknowledgment: Set
    .... .... 0... = Push: Not set
    .... .... .0.. = Reset: Not set
    .... .... ..1. = Syn: Set
    .... .... ...0 = Fin: Not set
```

Acknowledgment field is 1.

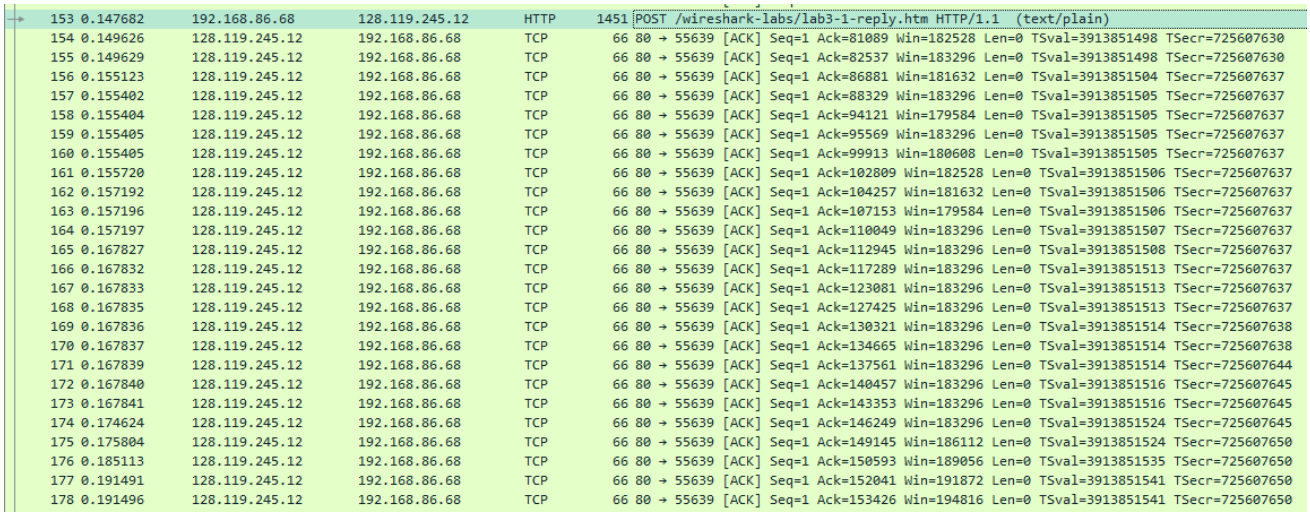
5. The sequence number is 152041.



152359 bytes.

```
Accept-Language: en-US,en;q=0.5\r\n
Accept-Encoding: gzip, deflate\r\n
Content-Type: multipart/form-data; boundary=---------------------------193950769437984536211154302391\r\n
Content-Length: 152359\r\n
Origin: http://gaia.cs.umass.edu\r\n
DNT: 1\r\n
```

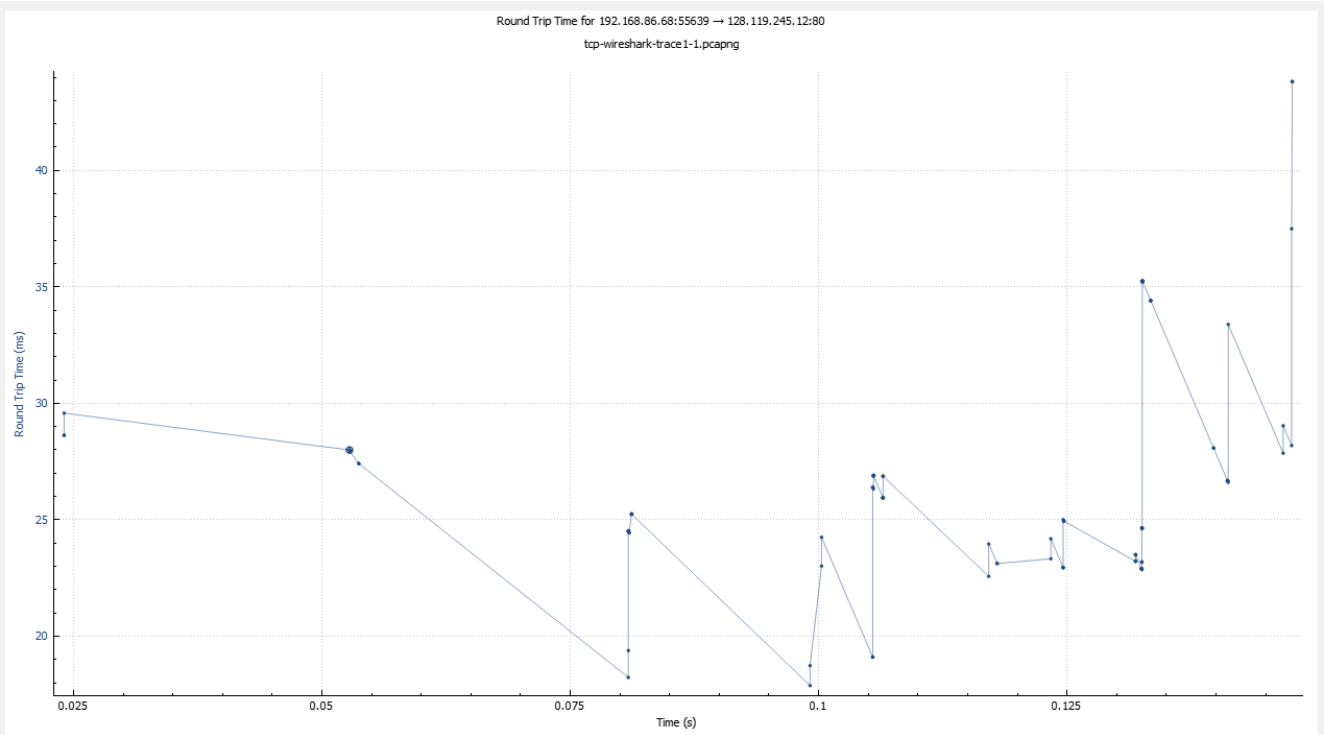
No, it is transferred in multiple segments.



6. At 0.149626, the first segment is sent.

   At 0.149629, the first ACK is received.

153 0.147682	192.168.86.68	128.119.245.12	HTTP	1451	POST /wireshark-l
154 0.149626	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK]
155 0.149629	128.119.245.12	192.168.86.68	TCP	66	80 → 55639 [ACK]



RTT for this first data-containing segment: It is about 28ms.

RTT value the second data-carrying TCP segment and its ACK: It is about 35ms.

7. HTTP Post segment is 1385 bytes.

   First segment is 52 bytes.

   Second segment is 52 bytes.

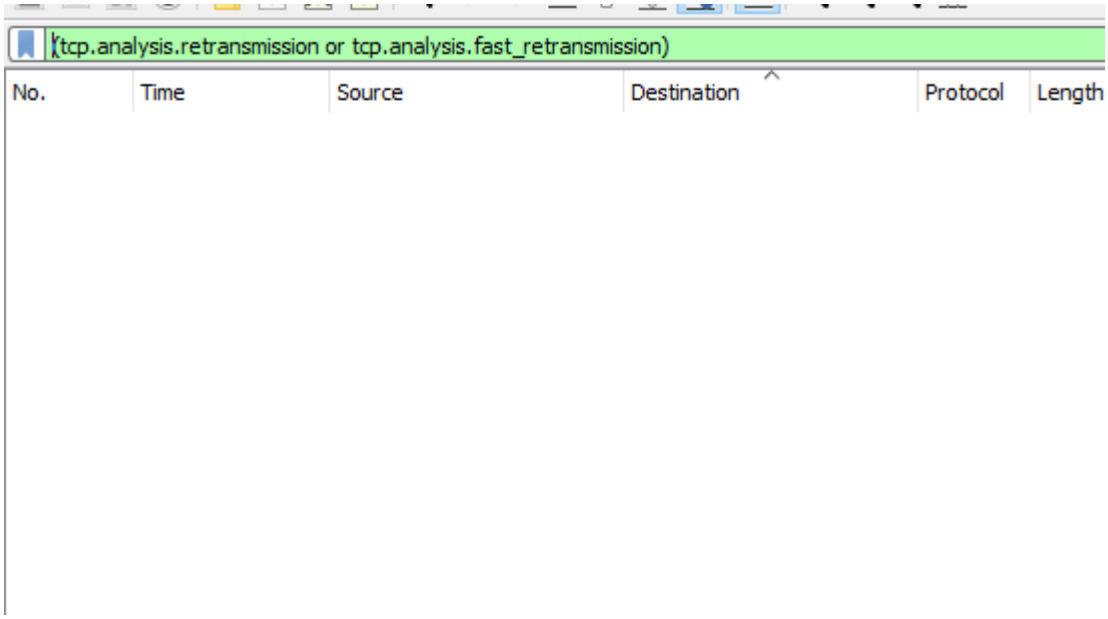
   Third segment is 52 bytes.

   Fourth segment is 52 bytes.

8. 182528 bytes.

```
Flags: 0x010 (ACK)
Window: 1426
[Calculated window size: 182528]
[Window size scaling factor: 128]
Checksum: 0xa454 [unverified]
```

9. There is no retransmission.



10.

	Ack sequence number	Ack data
ACK 1	1449	
ACK 2	2897	
ACK 3	4345	
ACK 4	5793	
ACK 5	7241	
ACK 6	8689	
ACK 7	10137	
ACK 8	11585	
ACK 9	13033	
ACK 10	14481	

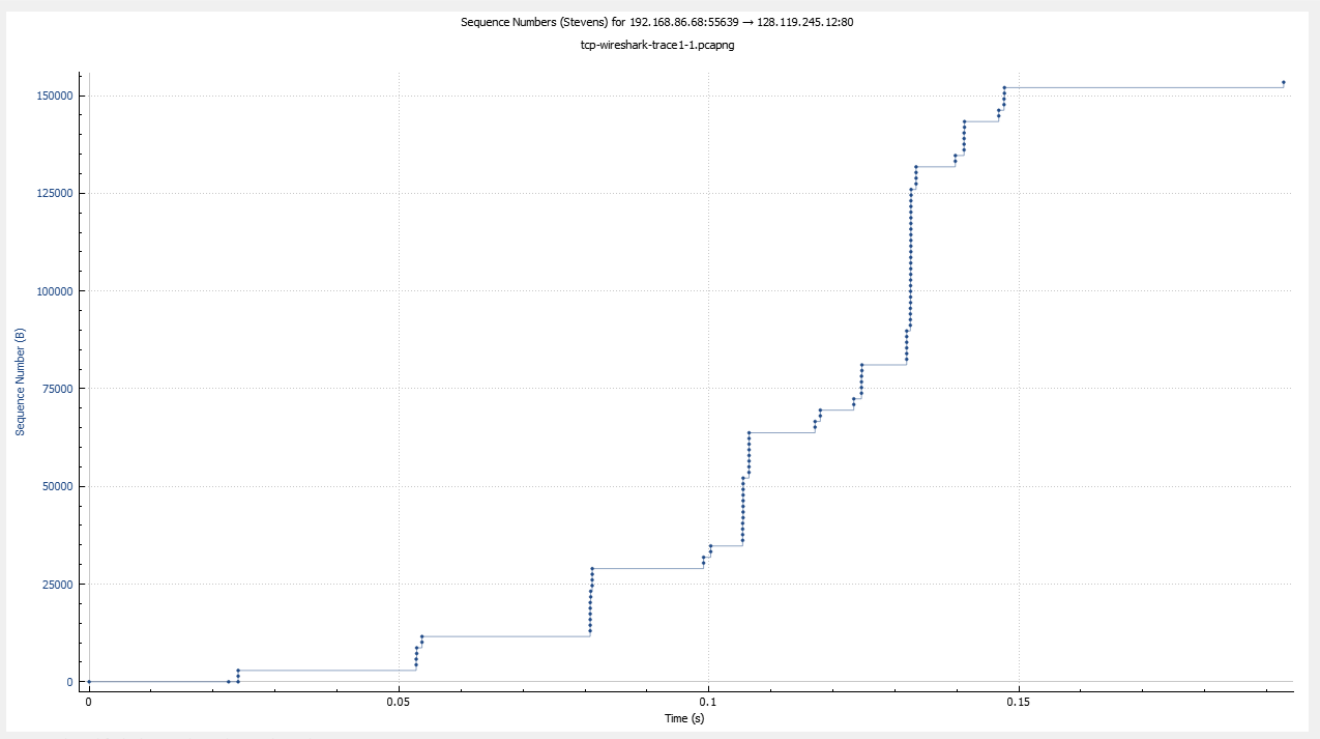
11.

Content length: 152359 bytes

Time passed: 0.192625-0.147682=0.044943 seconds

Throughput: Content length / Time passed = 3,310.60 Kb/s

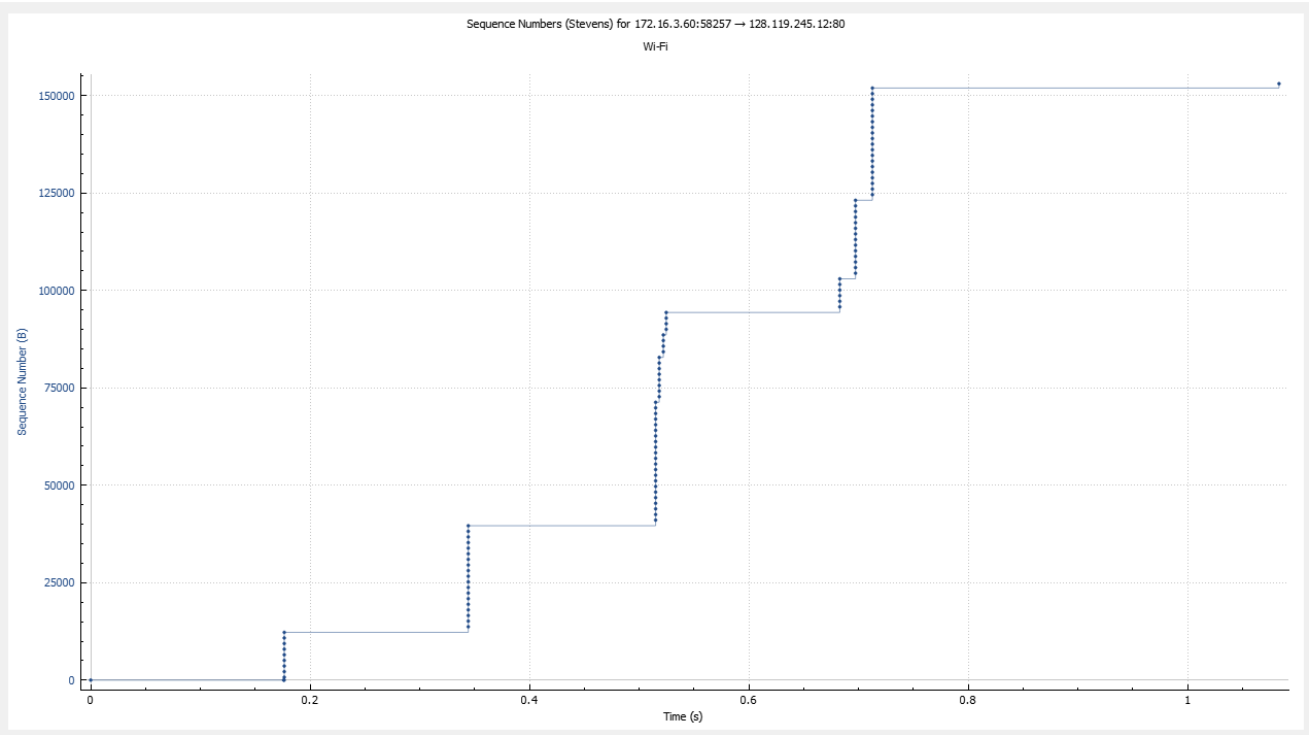
12.



13.

The period is getting smaller and smaller over time.

14.



The period is constant for the first couple of segments then it is getting smaller and smaller.